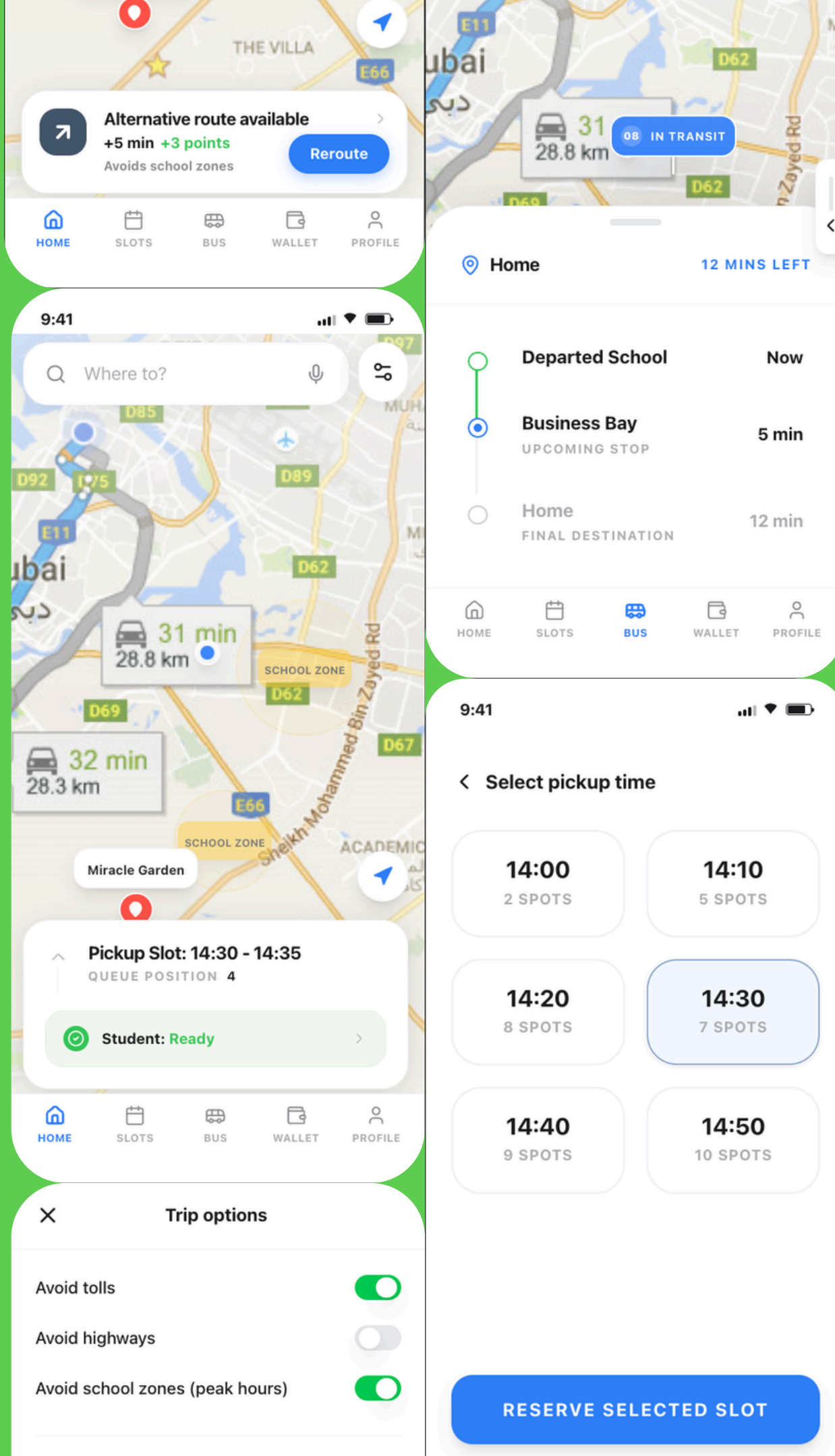




MASAR

LIVE MAP VIEW

BOOK SLOTS

STUDENT STATUS

ROUTE RECCOMENDATIONS

GPS PROXIMITY ALERT

REWARD POINTS

TRACK SCHOOL BUS

ZONE TOGGLE



BRIEF

Masar is an agentic AI traffic orchestration system designed to eliminate congestion in school zones. Unlike traditional solutions that rely on static infrastructure (more signs, more lanes), Masar manages the demand for road space dynamically.



PROBLEM

Why School Traffic Exists in Modern Cities:

School traffic isn't a random congestion problem, it's a predictable coordination failure.

1. Parents arrive early to avoid missing their child
2. Commuters choose the shortest route
3. Buses follow fixed schedules

When these decisions overlap in **Time & Space**, they collide.



THE SOLUTION

1

MASAR: A Virtual Traffic Controller

Coordinates parents, commuters, and buses before they collide, preventing bottlenecks rather than enforcing around them.

2

Keeping Traffic in Flow, Not in Conflict

Reroutes non-school traffic away from peak zones
Controls parent pickup flow to prevent lane blocking
Prioritizes buses without manual enforcement



STAKEHOLDERS

PICKUP GUARDIANS



MASAR controls pickup timing and queue order by dynamically assigning slots so parents don't arrive early or block lanes.

CIVILIAN COMMUTERS



MASAR reroutes non-school traffic away from school zones during peak hours to reduce unnecessary congestion.

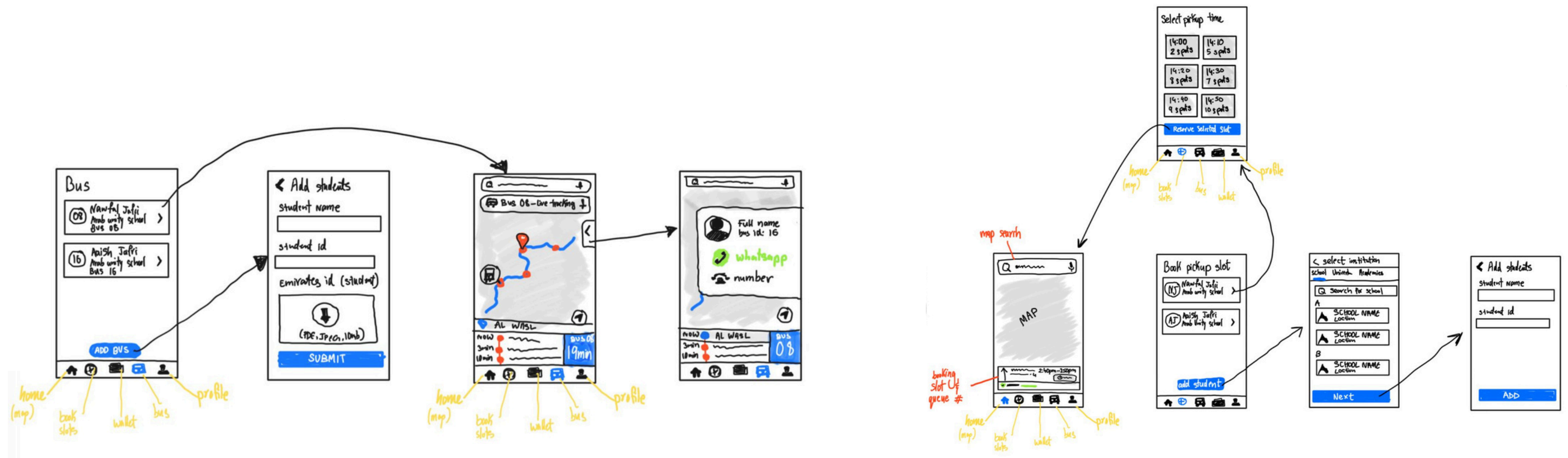
BUS DRIVERS



MASAR turns school buses into a live, traceable network—dynamically routed, centrally monitored, and auditable for safety and traffic control.



UX IDEATION



IN APP EXPERIENCE

9:41

Add students

Enter student information to track their bus

Student name

Enter student name

Student id

Enter student ID

Emirates id (student)

EMIRATES ID
784-XXXX-XXXXXXXX-X

784-1234-1234567-1

SUBMIT

HOME SLOTS **BUS** WALLET PROFILE

9:41

Bus

Track your school buses

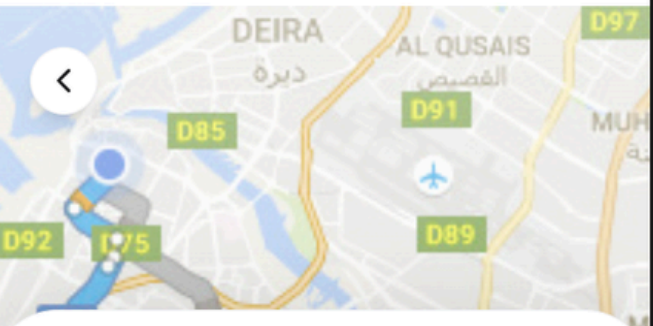
08 Hamdan Jafreer
Dubai Int. School

16 Amira Jafreer
Gems Dubai School

+ ADD BUS

HOME SLOTS **BUS** WALLET PROFILE

9:41



Home - Emirates Hills 10 MINS LEFT

School Zone Exit Now

Sheikh Zayed Road 3 min
UPCOMING STOP

Home - Emirates Hills 10 min

Al Barsha South Stop 14 min

Dubai Hills Estate 18 min

Meydan Road Stop 22 min
FINAL DESTINATION

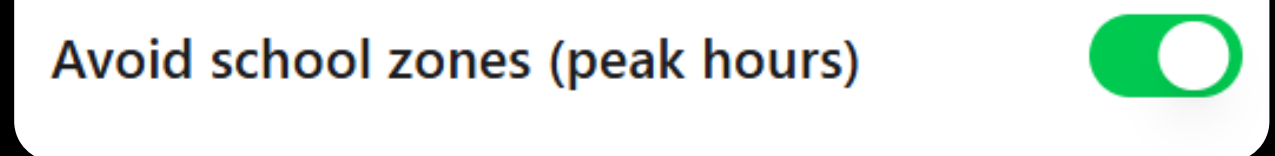
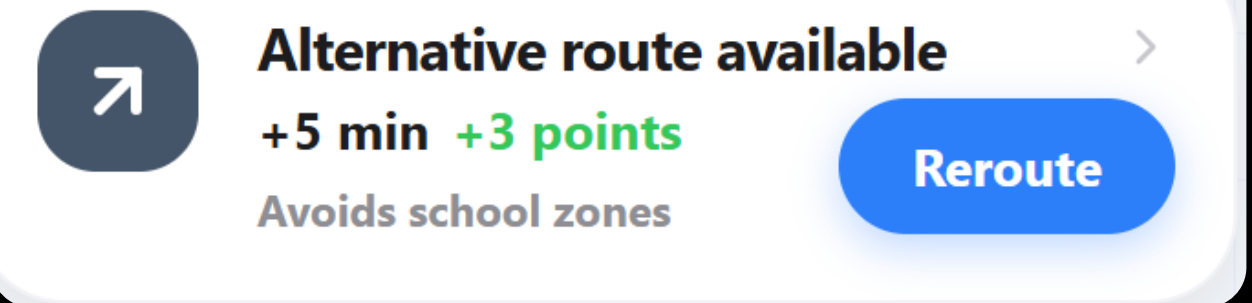
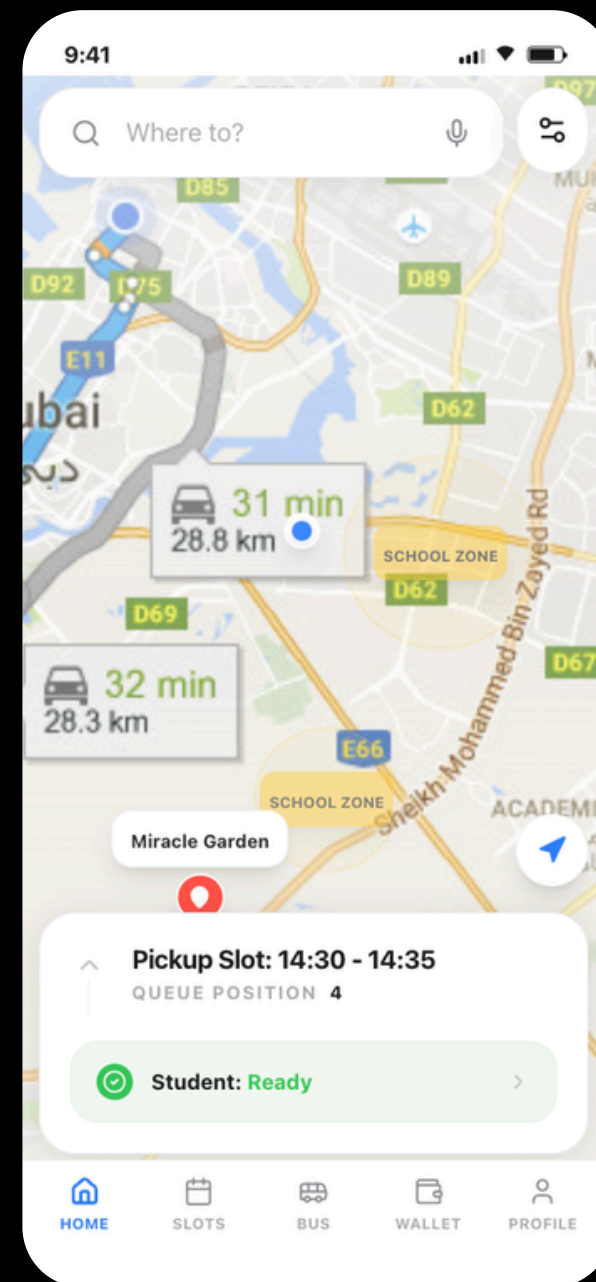
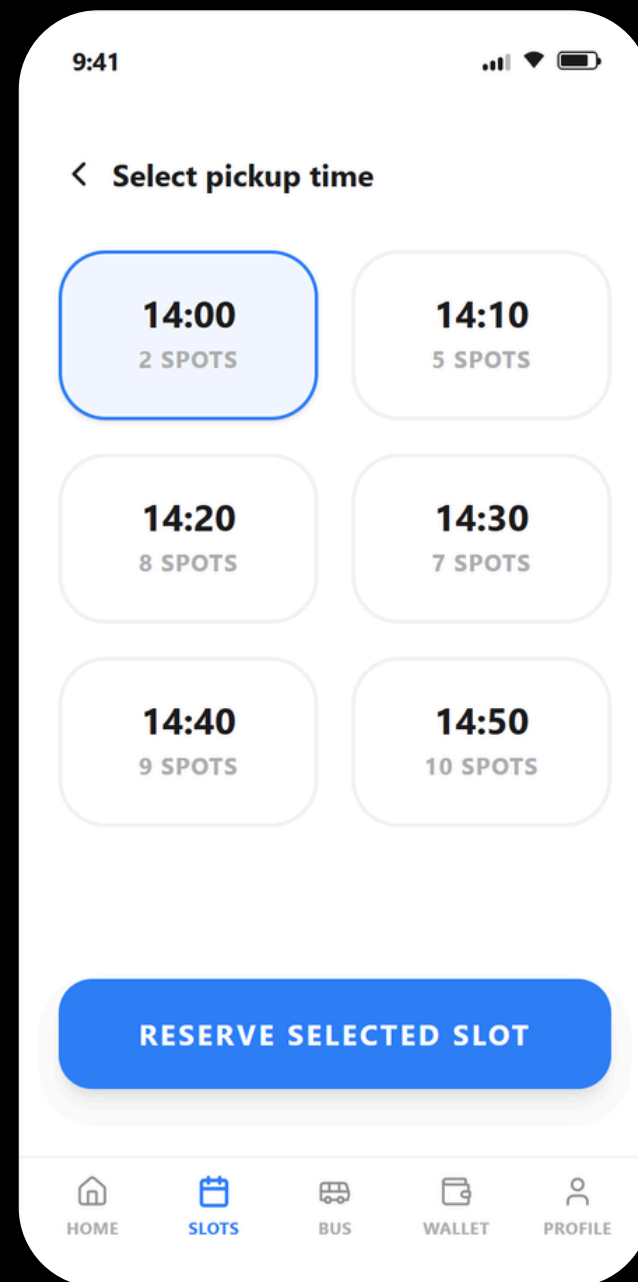
HOME SLOTS **BUS** WALLET PROFILE

Live school bus tracking for guardians, with centralized routing to stop data enabling the RTA to optimize bus operations, reduce unnecessary detours, and better management of traffic flow around school zones.



IN APP EXPERIENCE

Parents book a pickup time slot, and the system uses live location to notify the student when the car is approaching, reducing waiting time and keeping pickup lanes moving.



When toggled, the app reroutes non-school drivers away from school zones during peak hours, rewarding the choice of alternative routes to reduce congestion.



BACKEND LOGIC

- MASAR operates a **backend prediction service** that continuously learns when and where traffic congestion forms around school zones.
- An **LSTM model** is trained on historical traffic data (e.g., TomTom traffic datasets) to identify recurring high-congestion windows specific to each school's surrounding road network.
- The backend generates school-specific pickup time slots and dynamically determines the number of slots available per time window.
- These recommended time slots and capacities are packaged as a **JSON output** and sent to the main application which is used to:
 - Propose **congestion-aware routing** that avoids school zones for non-school trips
 - Enable structured pickup slot selection for parents, aligned with **predicted traffic conditions**



ADDITIONAL BENEFITS FOR RTA

- **Urban Planning:** school-run data to identify congestion patterns and support long-term planning.
 - **Safety & Accountability:** centralized logs enable traceability of student during emergencies.
 - **Adoption & Insights:** Integrating MASAR into S'hail increases app adoption while expanding the RTA's mobility and behavioral data coverage.
-

SAFETY AND VERIFICATION

- Users authenticate using **UAE Pass** to access MASAR features
- Adding a student for pickup requires a **valid Student ID**
- Access to school bus tracking requires **Emirates ID** verification
- Multi-level verification ensures only **authorized** guardians can manage student's data



